

DDL Commands – CREATE, ALTER, DROP

1) Create a table name STUDENT with following structure.

#	Column Name	Description	Data Type
1	RegNo	Registration Number	NUMBER(3)
2	Name	Student Name	VARCHAR(15)
3	Gender	Gender of the student	CHAR(1)
4	DOB	Date of Birth	DATE
5	MobileNo	Mobile Number	NUMBER(10)
6	City	Location of stay	VARCHAR(15)

```
mysql> create table student(Regno int(3),Name char(15),gender char(1),Dob int(10),mobilenno int(10),city char(10));
Query OK, 0 rows affected (0.14 sec)

mysql> desc student;
+-----+-----+-----+-----+-----+-----+
| Field | Type  | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Regno | int(3) | YES  |     | NULL    |       |
| Name  | char(15) | YES  |     | NULL    |       |
| gender | char(1) | YES  |     | NULL    |       |
| Dob   | int(10) | YES  |     | NULL    |       |
| mobilenno | int(10) | YES  |     | NULL    |       |
| city  | char(10) | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

2) Create a table name FACULTY with following structure.

#	Column Name	Description	Data Type
1	FacNo	Faculty Identifier	VARCHAR(4)
2	FacName	Faculty Name	VARCHAR(15)
3	Gender	Gender of faculty	CHAR(1)
4	DOB	Date of Birth	DATE
5	DOJ	Date of Join	DATE
6	MobileNo	Mobile Number	NUMBER(10)

```
mysql> create table faculty(Facno int(3),FacName char(15),gender char(1),Dob int(10),mobilen no int(10),DOJ int(10));
Query OK, 0 rows affected (0.06 sec)

mysql> desc faculty;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Facno | int(3) | YES | | NULL | |
| FacName | char(15) | YES | | NULL | |
| gender | char(1) | YES | | NULL | |
| Dob | int(10) | YES | | NULL | |
| mobilen o | int(10) | YES | | NULL | |
| DOJ | int(10) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

3) Create a table name DEPARTMENT with following structure.

#	Column Name	Description	Data Type
1	DeptNo	Department Identifier	VARCHAR(4)
2	DeptName	Department Name	VARCHAR(15)
3	DeptHead	Department Head	VARCHAR(4)

```
mysql> create table department(deptno int(10),deptname char(10),depthead char(10));
Query OK, 0 rows affected (0.11 sec)

mysql> desc department;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| deptno | int(10) | YES | | NULL | |
| deptname | char(10) | YES | | NULL | |
| depthead | char(10) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

4) Create a table name COURSE with following structure.

#	Column Name	Description	Data Type
1	CourseNo	Course Identifier	VARCHAR(3)
2	CourseDesc	Course Description	VARCHAR(14)
3	CourseType	Course Type	CHAR(1)
4	SemNo	Semester Number	CHAR(1)
5	HallNo	Hall Number	VARCHAR(4)
6	FacNo	Faculty Identifier	VARCHAR(4)

```
mysql> create table course(courseno int(3),coursedesc char(15),coursetype char(1),semno int(10),hallno int(10),Facno int(10));
Query OK, 0 rows affected (0.09 sec)

mysql> desc course;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| courseno | int(3) | YES | | NULL | |
| coursedesc | char(15) | YES | | NULL | |
| coursetype | char(1) | YES | | NULL | |
| semno | int(10) | YES | | NULL | |
| hallno | int(10) | YES | | NULL | |
| Facno | int(10) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

5) Modify the table FACULTY by adding a column name Dept of datatype VARCHAR(10)

```
mysql> alter table faculty add dept char(10);
Query OK, 2 rows affected (0.09 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> desc faculty;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Facno | int(3) | NO | PRI | NULL | |
| FacName | char(15) | YES | | NULL | |
| gender | char(1) | YES | | NULL | |
| Dob | int(10) | YES | | NULL | |
| mobileno | int(10) | YES | | NULL | |
| DOJ | int(10) | YES | | NULL | |
| dept | char(10) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)
```

DDL Commands with Constraints – PRIMARY, FOREIGN KEY, UNIQUE, CHECK

1) Alter the table STUDENT2 with following structure.

#	Column Name	Constraints
1	RegNo	PRIMARY KEY

2) Alter the table name FACULTY with following structure.

#	Column Name	Constraints
1	FacNo	PRIMARY KEY
2	Gender	CHECK 'M' or 'F'

3) Alter the table name DEPARTMENT with following structure.

#	Column Name	Constraints
1	DeptNo	PRIMARY KEY

4) Alter the table name COURSE with following structure.

#	Column Name	Constraints
1	CourseNo	PRIMARY KEY
2	SemNo	1 to 6

5) Alter the table name STUDENT2 with following structure.

#	Column Name	Constraints
1	S1Name	UNIQUE KEY

6) After the STUDENT2 & STUDENT3 tables are successfully created, test if you can add a constraint FOREIGN KEY to the RegNo of this table.

OUTPUTS:

1)

```
mysql> Alter table student2 ADD PRIMARY KEY(RegNo);
Query OK, 0 rows affected (0.05 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> desc student2;
```

Field	Type	Null	Key	Default	Extra
RegNo	int	NO	PRI	NULL	
S1Name	varchar(15)	YES		NULL	
Age	char(2)	YES		NULL	
MobileNo	int	YES		NULL	
address	varchar(15)	YES		NULL	

5 rows in set (0.01 sec)

2)

```
mysql> alter table faculty add primary key(facno);
Query OK, 0 rows affected (0.07 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> alter table faculty add check(gender='M'or'F');
Query OK, 0 rows affected (0.06 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> desc faculty;
```

Field	Type	Null	Key	Default	Extra
Facno	int(3)	NO	PRI	NULL	
FacName	char(15)	YES		NULL	
gender	char(1)	YES		NULL	
Dob	int(10)	YES		NULL	
mobilenno	int(10)	YES		NULL	
DOJ	int(10)	YES		NULL	

6 rows in set (0.00 sec)

3)

```
mysql> alter table department add primary key(deptno);
Query OK, 0 rows affected (0.11 sec)
```

```
mysql> desc department;
```

Field	Type	Null	Key	Default	Extra
deptno	int(10)	NO	PRI	NULL	
deptname	char(10)	YES		NULL	
depthead	char(10)	YES		NULL	

3 rows in set (0.00 sec)

4)


```
mysql> alter table course add primary key(courseno);
Query OK, 0 rows affected (0.07 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> alter table course add check(semno>=1&&semno<=6);
Query OK, 0 rows affected (0.11 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc course;
```

Field	Type	Null	Key	Default	Extra
courseno	int(3)	NO	PRI	NULL	
coursedes	char(15)	YES		NULL	
coursetype	char(1)	YES		NULL	
semno	int(10)	YES		NULL	
hallno	int(10)	YES		NULL	
Facno	int(10)	YES		NULL	

```
6 rows in set (0.00 sec)
```

5)

```
mysql> Alter table student2 ADD UNIQUE KEY(S1Name);
Query OK, 0 rows affected (0.07 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc student2;
```

Field	Type	Null	Key	Default	Extra
RegNo	int	NO	PRI	NULL	
S1Name	varchar(15)	YES	UNI	NULL	
Age	char(2)	YES		NULL	
MobileNo	int	YES		NULL	
address	varchar(15)	YES		NULL	

```
5 rows in set (0.00 sec)
```

6)

```
mysql> ALTER TABLE student1 modify address int not null;
Query OK, 0 rows affected (0.07 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc student1;
```

Field	Type	Null	Key	Default	Extra
RegNo	int	NO	PRI	NULL	
S1Name	varchar(15)	YES		NULL	
Age	char(2)	YES	UNI	NULL	
DOB	int	YES		NULL	
address	int	NO		NULL	

```
5 rows in set (0.00 sec)
```

DML Commands – INSERT, SELECT

Questions:

1. Populate all the five tables with your own data.

```
mysql> insert into student values
-> (19221101,'Jayashree','F','2005-06-15',9876543210,'Chennai'),
-> (19221102,'Priya','F','2005-09-20',9876543211,'Coimbatore'),
-> (19221103,'Kavin','M','2004-12-10',9876543212,'Madurai'),
-> (19221104,'Arun','M','2005-03-18',9876543213,'Salem'),
-> (19221105,'Divya','F','2005-11-25',9876543214,'Trichy');
Query OK, 5 rows affected (0.02 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> select * from student;
```

Reg_no	Name	Gender	DOB	Mobile_no	City
19221101	Jayashree	F	2005-06-15	9876543210	Chennai
19221102	Priya	F	2005-09-20	9876543211	Coimbatore
19221103	Kavin	M	2004-12-10	9876543212	Madurai
19221104	Arun	M	2005-03-18	9876543213	Salem
19221105	Divya	F	2005-11-25	9876543214	Trichy

```
5 rows in set (0.00 sec)

mysql> _
```

```
mysql> insert into course values
-> ('CS101','Programming in C','Mandatory',1,'H001','F001'),
-> ('CS102','Database Management','Mandatory',2,'H002','F002'),
-> ('CS103','Data Structures','Mandatory',3,'H003','F003'),
-> ('CS104','Computer Networks','Elective',4,'H004','F004'),
-> ('CS105','Artificial Intelligence','Elective',5,'H005','F005');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> select * from course;
+-----+-----+-----+-----+-----+-----+
| CourseNo | CourseDesc | CourseType | SemNo | HallNo | FacNo |
+-----+-----+-----+-----+-----+-----+
| CS101 | Programming in C | Mandatory | 1 | H001 | F001 |
| CS102 | Database Management | Mandatory | 2 | H002 | F002 |
| CS103 | Data Structures | Mandatory | 3 | H003 | F003 |
| CS104 | Computer Networks | Elective | 4 | H004 | F004 |
| CS105 | Artificial Intelligence | Elective | 5 | H005 | F005 |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql>
```

```
mysql> insert into department values
-> (1,'Computer Science','Dr.Meena'),
-> (2,'Electronics','Dr.Ramesh'),
-> (3,'Electrical','Dr.Suresh'),
-> (4,'Information Technology','Dr.Anitha'),
-> (5,'Mechanical','Dr.Kumar');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> select * from department;
+-----+-----+-----+
| Deptno | Deptname | Depthead |
+-----+-----+-----+
| 1 | Computer Science | Dr.Meena |
| 2 | Electronics | Dr.Ramesh |
| 3 | Electrical | Dr.Suresh |
| 4 | Information Technology | Dr.Anitha |
| 5 | Mechanical | Dr.Kumar |
+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> _
```

```
mysql> insert into faculty values
-> ('F001','Meena','F','1985-04-10','2010-06-12',9876500001,'CSE','N'),
-> ('F002','Ramesh','M','1980-08-15','2008-05-20',9876500002,'ECE','Y'),
-> ('F003','Suresh','M','1983-12-18','2012-07-25',9876500003,'EEE','N'),
-> ('F004','Anitha','F','1988-03-14','2015-09-10',9876500004,'IT','Y'),
-> ('F005','Kumar','M','1982-11-28','2009-01-05',9876500005,'MECH','N');
Query OK, 5 rows affected (0.02 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> select * from faculty;
+-----+-----+-----+-----+-----+-----+-----+
| FacNo | Facname | Gender | DOB | DOJ | Mobileno | Deptno | Resigned |
+-----+-----+-----+-----+-----+-----+-----+
| F001 | Meena | F | 1985-04-10 | 2010-06-12 | 9876500001 | CSE | N |
| F002 | Ramesh | M | 1980-08-15 | 2008-05-20 | 9876500002 | ECE | Y |
| F003 | Suresh | M | 1983-12-18 | 2012-07-25 | 9876500003 | EEE | N |
| F004 | Anitha | F | 1988-03-14 | 2015-09-10 | 9876500004 | IT | Y |
| F005 | Kumar | M | 1982-11-28 | 2009-01-05 | 9876500005 | MECH | N |
+-----+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> _
```



```
mysql> insert into hall values
-> ('H001','Block A',60),
-> ('H002','Block B',55),
-> ('H003','Block C',50),
-> ('H004','Block D',45),
-> ('H005','Block E',70);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> select * from hall;
+-----+-----+-----+
| Hallno | Hallname | Capacity |
+-----+-----+-----+
| H001   | Block A  | 60      |
| H002   | Block B  | 55      |
| H003   | Block C  | 50      |
| H004   | Block D  | 45      |
| H005   | Block E  | 70      |
+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> _
```

2) View all the records from the five tables.

```
mysql> select * from student;
+-----+-----+-----+-----+-----+-----+
| Reg_no | Name      | Gender | DOB       | Mobile_no | City      |
+-----+-----+-----+-----+-----+-----+
| 19221101 | Jayashree | F      | 2005-06-15 | 9876543210 | Chennai  |
| 19221102 | Priya     | F      | 2005-09-20 | 9876543211 | Coimbatore |
| 19221103 | Kavin     | M      | 2004-12-10 | 9876543212 | Madurai  |
| 19221104 | Arun      | M      | 2005-03-18 | 9876543213 | Salem   |
| 19221105 | Divya     | F      | 2005-11-25 | 9876543214 | Trichy   |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> _
```

```
mysql> select * from faculty;
```

Facno	Facname	Gender	DOB	DOJ	Mobileno	Deptno	Resigned
F001	Meena	F	1985-04-10	2010-06-12	9876500001	CSE	N
F002	Ramesh	M	1980-08-15	2008-05-20	9876500002	ECE	Y
F003	Suresh	M	1983-12-18	2012-07-25	9876500003	EEE	N
F004	Anitha	F	1988-03-14	2015-09-10	9876500004	IT	Y
F005	Kumar	M	1982-11-28	2009-01-05	9876500005	MECH	N

```
5 rows in set (0.00 sec)
```

```
mysql> _
```

```
mysql> select * from course;
```

Courseno	Coursedesc	Coursetype	Semno	Hallno	Facno
CS101	Programming in C	Mandatory	1	H001	F001
CS102	Database Management	Mandatory	2	H002	F002
CS103	Data Structures	Mandatory	3	H003	F003
CS104	Computer Networks	Elective	4	H004	F004
CS105	Artificial Intelligence	Elective	5	H005	F005

```
5 rows in set (0.00 sec)
```

```
mysql> _
```

```
mysql> select * from department;
```

Deptno	Deptname	Depthead
1	Computer Science	Dr.Meena
2	Electronics	Dr.Ramesh
3	Electrical	Dr.Suresh
4	Information Technology	Dr.Anitha
5	Mechanical	Dr.Kumar

```
5 rows in set (0.00 sec)
```

```
mysql> _
```

```
mysql> select * from hall;
+-----+-----+-----+
| Hallno | Hallname | Capacity |
+-----+-----+-----+
| H001   | Block A  | 60       |
| H002   | Block B  | 55       |
| H003   | Block C  | 50       |
| H004   | Block D  | 45       |
| H005   | Block E  | 70       |
+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> _
```

EXPERIMENT - 3

DML Commands with Constraints –UPDATE, DELETE

Questions:

1. Update the value of student name whose register number is '191711342'

```
mysql> UPDATE student
-> SET Name = 'Jayashree'
-> WHERE Reg_no = 192524146;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM student
-> WHERE Reg_no = 192524146;
+-----+-----+-----+-----+-----+-----+
| Reg_no | Name      | Gender | DOB       | Mobile_no | City    |
+-----+-----+-----+-----+-----+-----+
| 192524146 | Jayashree | F      | 2007-10-20 | 9876543210 | Chennai |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> _
```

2. Delete the record in the table FACULTY, who resigned her job.

```
mysql> delete from faculty where facname='Mohan';  
Query OK, 1 row affected (0.02 sec)
```

```
mysql> select * from faculty;
```

facno	facname	Gender	DOB	DOJ	Mobileno	deptno	resigned
F1922113	ramesh	m	2005-03-14	2014-12-20	998263548	mnr	y
F1922115	san	m	2007-05-17	2016-09-20	998269245	hyt	y
F1922112	raju	m	2005-02-13	2009-06-25	90865894	mgr	y

3 rows in set (0.00 sec)

```
mysql> █
```

3. Modify the age for the faculty whose name is 'mohan' with a value '59'.

```
mysql> select * from faculty;
```

facno	facname	Gender	DOB	DOJ	Mobileno	deptno	resigned	age
F1922113	ramesh	m	2005-03-14	2014-12-20	998263548	mnr	y	45
F1922115	san	m	2007-05-17	2016-09-20	998269245	hyt	y	68
1191151	mohan	m	2004-01-12	2010-05-23	990826973	mtr	y	59
1f1922112	raju	m	2005-02-13	2009-06-25	90865894	mgr	y	45

4 rows in set (0.00 sec)

```
mysql> █
```


4. Remove all faculty who are having over 65 years .

```
mysql> delete from faculty where age > 65;
```

```
Query OK, 1 row affected (0.02 sec)
```

```
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> select * from faculty;
```

facno	facname	Gender	DOB	DOJ	Mobileno	deptno	resigned	age
F1922113	ramesh	m	2005-03-14	2014-12-20	998263548	mnr	y	45
1191151	mohan	m	2004-01-12	2010-05-23	990826973	mtr	y	59
1f1922112	raju	m	2005-02-13	2009-06-25	90865894	mgr	y	45

```
3 rows in set (0.00 sec)
```

```
mysql> █
```